

Assignment 1

Assignment 1 should be handed in before the **9th** of December 2025. It consists of four different problems. The **code should be handed in** as well as a small text with **a minimum of 500 words** explaining the approaches and the value these could have in a musical setting. The assignment is worth 30% of the final grade.

1. Included with the assignment is a data file, *cities.csv* that contains information about world cities and various numbers that relate to them. You should use this and implement a **Sonification process** that highlights some characteristics of the data in the file. There should be a **written reflection** on the data chosen, what makes it interesting and how its properties are reflected in the sonification procedure. The other requirement is that all of the sound produced by the process should be based on **waveform generating processes** (discussed in the class Microsound 2). You can also propose another dataset to sonify, including those discussed in class, but please consult with me first.

The written text should reflect on the sonification strategy, the choices made and how they relate to the input data.

2. Stochastic Synthesis. By using dynamic waveforms that continuously change, many approaches to microsound can be developed. Create a short etude lasting at least 1 minute, with a constantly evolving waveform throughout.

The written text should reflect on the characteristics of how the waveform is changing and the sonorities it creates.

3. Chaos. Chaotic functions can produce varying shapes that vary over time and are sensitive to initial conditions. Implement **two layers** of sound using chaotic functions. The two layers should follow distinct paths and serve

different functions in the musical output. Finally, design at least three **configurations** of the layers that can be activated in a different temporal order based on input parameters.

The written text should reflect on the characteristics of the functions and what different behaviours they are capable of generating.

4. Create a short **étude** of about *three* or more minutes that explores **granular synthesis**. There should be some usage of *granular synthesis*, but there could also be pulsar or waveset techniques used. Try to explore how ideas, shapes or movements can be used in different ways across various *time scales*.

The written text here should reflect on how composing can be seen as taking place on the sound-itself as well as for a larger form. Also explain how the form came about as well as the material creation.